

Oscar Antonio Juárez Jiménez

AI / Machine Learning Engineer

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AI Engineer (B.Eng., IPN-ESCOM) with 2 years of experience in banking and enterprise environments building and shipping end-to-end data, Machine Learning, and LLM systems: PySpark ETL pipelines on AWS, containerized services on Docker and Kubernetes, and CI/CD-driven releases. Focused on Large Language Models, RAG, and distributed data architectures, with a track record of turning ambiguous business problems into measurable outcomes alongside distributed teams.

Professional Experience

Bluetab, an IBM Company — client: BBVA

Jul 2025 – Present

Data Engineer

Mexico City (Remote)

- Cut end-to-end data preparation time by **40%** for analytics and ML teams by designing and taking to production PySpark ETL pipelines that integrate **10 source systems** into the bank's core data platform.
- Reduced Spark job cost and scanned volume by **22%** — measured as cluster time per job — by restructuring datasets and tables on AWS S3 with partitioning and schema governance.
- Unblocked **10 ML use cases** in the structural risk division by translating feature requirements into production-grade data products adopted by modeling teams, working directly with data scientists and risk analysts.

Grupo Salinas

Feb – Jul 2025

AI Engineer, Innovation & Development (Intern)

Mexico City

- Halved manual handling time for administrative workflows (**50%** lower average resolution time per request) by deploying LLM-based agents and chatbots built with LangChain and the OpenAI APIs.
- Shortened prototype-to-validation cycles by delivering interactive prototypes and dynamic landing pages under a Design Thinking process, letting business units test AI concepts before committing engineering effort.

AI Projects

3D Object Segmentation in Indoor Scenes (NeRF + Diffusion)

Jun 2024 – Aug 2025

B.Eng. Thesis — PyTorch, NeRF, Stable Diffusion

IPN-ESCOM

- Improved segmentation accuracy by **30%** over the U-Net baseline, measured by mIoU across **200 annotated indoor scenes**, by combining classical ML classifiers with a U-Net architecture in PyTorch.
- Achieved photorealistic 3D scene editing by pairing NeRF spatial reconstruction with diffusion-based texture synthesis (Stable Diffusion), and documented the architecture and evaluation protocol for full reproducibility.

RAG Chatbot over Mexico's Official Gazette (DOF)

Jan – Jun 2024

LangChain, LangServe, Pinecone, Docker

Personal project

- Cut legal document search time from **~1 hour to 90 seconds per query** (**~97%** faster) with a Retrieval-Augmented Generation system that answers and summarizes natural-language questions over the DOF.
- Shipped the system as a containerized MCP server, using LangChain and LangServe for orchestration and Pinecone as the vector database, with Docker for reproducible builds and deployments.

Technical Skills

Machine Learning & AI: Deep learning, neural networks, CNN/U-Net, computer vision, NLP, Large Language Models, Retrieval-Augmented Generation (RAG), prompt engineering, model evaluation and error analysis | PyTorch, TensorFlow, scikit-learn, LangChain, LangServe, Pinecone, Stable Diffusion

Cloud, MLOps & Infrastructure: AWS (S3, EMR, Glue, IAM), Google Cloud Platform, Azure, Docker, Kubernetes, CI/CD, Git/GitHub, MLflow, model deployment and monitoring

Data Engineering: Apache Spark / PySpark, distributed clusters, ETL design, SQL and query optimization, data modeling, schema and partition governance, vector databases

Programming: Python, Java, SQL, Scala, C/C++, JavaScript/TypeScript, Go, CUDA, Bash

Education

Instituto Politécnico Nacional (IPN) — ESCOM

Aug 2021 – Jul 2025

B.Eng. in Artificial Intelligence — GPA 9.5/10 ($\approx 3.8/4.0$)

Mexico City, Mexico

Relevant coursework: Machine Learning, Neural Networks and Deep Learning (foundations and advanced), Natural Language Technologies, Digital Image Processing, Parallel Computing, Bio-inspired Algorithms.

Certifications & Languages

Certifications: AWS Certified Data Engineer; AWS Certified Cloud Practitioner; AWS Certified AI Practitioner; Anthropic Academy (full course track); Google Cloud Computing Foundations; Google Cloud GenAI Skill Badges; Microsoft Learn AI Skills Challenge (Azure); SAFe Scrum Master (Scaled Agile)

Languages: Spanish — native | English — C1 (professional working proficiency)